

Exercise #8.6

Q#1: Show that the shortest distance between the lines $x + a = 2y = -12z$ and $x = y + 2a = 6(z - a)$ is $2a$.

Solution: Given lines are

$$\left. \begin{aligned} x + a &= 2y = -12z \\ x &= y + 2a = 6(z - a) \end{aligned} \right\}$$

$$\left[\frac{x+a}{1} = \frac{y}{\frac{1}{2}} = \frac{z}{-\frac{1}{12}} \right]$$

$$\frac{x+a}{12} = \frac{y}{6} = \frac{z}{-1} \quad \text{----- (1)}$$

$$x - y - 2a = 0 = x - 6z + 6a \quad \text{----- (2)}$$

Now equation of a plane through line (2) is

$$(x - y - 2a) + k(x - 6z + 6a) = 0 \Rightarrow (1+k)x - y - 6kz - 2a + 6ka = 0$$

Now direction ratios of normal vector of this plane are $1+k, -1, -6k$

$$\text{Then } 12(1+k) + 6(-1) - 1(-6k) = 0 \Rightarrow 12 + 12k - 6 + 6k = 0$$

$$18k + 6 = 0 \Rightarrow 3k + 1 = 0 \Rightarrow 3k = -1 \Rightarrow k = -\frac{1}{3}$$

Putting in equation of plane

$$(x - y - 2a) - \frac{1}{3}(x - 6z + 6a) = 0$$

$$3x - 3y - 6a - x + 6z - 6a = 0$$

$2x - 3y + 6z - 12a = 0$ is required plane through line (2) and parallel to line (1)

Let d be the required shortest distance then

$$d = \text{Distance of point } (-a, 0, 0) \text{ from plane } 2x - 3y + 6z - 12a = 0$$

$$d = \frac{|2(-a) - 0 + 0 - 12a|}{\sqrt{(2)^2 + (-3)^2 + (6)^2}} \Rightarrow d = \frac{|-2a - 12a|}{\sqrt{4 + 9 + 36}} \Rightarrow d = \frac{|-14a|}{\sqrt{49}} \Rightarrow d = \frac{14a}{7} \Rightarrow d = 2a$$

Q#2: Find the shortest distance between the x-axis and the straight line

$$ax + by + cz + d = 0 = a'x + b'y + c'z + d'$$

Solution: We know that equation of x-axis in symmetric form is

$$\frac{x}{1} = \frac{y}{0} = \frac{z}{0} \quad \text{----- (1)}$$

Now given line is

$$ax + by + cz + d = 0 = a'x + b'y + c'z + d' \quad \text{----- (2)}$$

Then equation of a plane containing this line is

$$(ax + by + cz + d) + k(a'x + b'y + c'z + d') = 0$$

$$\Rightarrow (a + ka')x + (b + kb')y + (c + kc')z + (d + kd') = 0$$

As this plane is parallel to x-axis

$$\text{So } 1(a + ka') = 0 \Rightarrow a + ka' = 0 \quad ka' = -a \quad k = -\frac{a}{a'}$$

Putting in equation of plane

$$(ax + by + cz + d) - \frac{a}{a'}(a'x + b'y + c'z + d') = 0$$

$$aa'x + a'by + a'cz + a'd - aa'x - ab'y - ac'z - ad' = 0$$

$$(a'b - ab')y + (a'c - ac')z + (a'd - ad') = 0 \text{ is equation of plane containing line (2)}$$

Let d' be the required shortest distance then

$d' = \text{Distance of point } (0,0,0) \text{ from plane}$

$$d' = \frac{|(a'b - ab')(0) + (a'c - ac')(0) + (a'd - ad')|}{\sqrt{(a'b - ab')^2 + (a'c - ac')^2}}$$

$$\Rightarrow d' = \frac{a'd - ad'}{\sqrt{(a'b - ab')^2 + (a'c - ac')^2}} \text{ required distance}$$

Q#3: Show that the shortest distance between the straight lines

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} \quad \text{and} \quad \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5} \text{ is } \frac{1}{\sqrt{6}}$$

and equations of the straight line perpendicular to both are $11x + 2y - 7z + 6 = 0 = 7x + y - 5z + 7$.

Solution: Given lines are

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} \quad \text{--- (1)}$$

$$\frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5} \quad \text{--- (2)}$$

A point on line (1) is $A(1,2,3)$

A point on line (2) is $B(2,4,5)$

$$\overrightarrow{AB} = (2-1)\hat{i} + (4-2)\hat{j} + (5-3)\hat{k} \Rightarrow \overrightarrow{AB} = \hat{i} + 2\hat{j} + 2\hat{k}$$

Here direction ratios of line (1) are 2,3,4 & direction ratios of line (2) are 3,4,5

Let $\vec{u}$ be a vector perpendicular to both given lines then

$$\vec{u} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}$$

Expanding from R_1

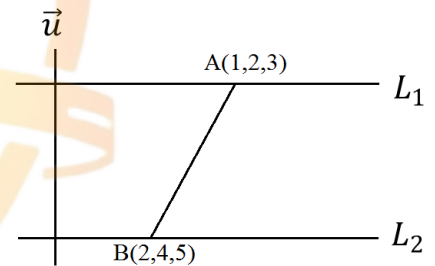
$$\vec{u} = (15 - 16)\hat{i} - (10 - 12)\hat{j} + (8 - 9)\hat{k}$$

$$\Rightarrow \vec{u} = -\hat{i} + 2\hat{j} - \hat{k}$$

Let d be the required shortest distance between lines then

$$d = \frac{|\overrightarrow{AB} \cdot \vec{u}|}{|\vec{u}|} \Rightarrow d = \frac{(\hat{i} + 2\hat{j} + 2\hat{k}) \cdot (-\hat{i} + 2\hat{j} - \hat{k})}{\sqrt{1+4+1}} = \frac{1(-1) + 2(2) + 2(-1)}{\sqrt{6}} \Rightarrow d = \frac{-1+4-2}{\sqrt{6}} \Rightarrow d = \frac{1}{\sqrt{6}}$$

is required distance



Now equations of line perpendicular to both given lines is

$$\begin{vmatrix} x-1 & y-2 & z-3 \\ 2 & 3 & 4 \\ -1 & 2 & -1 \end{vmatrix} = 0 = \begin{vmatrix} x-2 & y-4 & z-5 \\ 3 & 4 & 5 \\ -1 & 2 & -1 \end{vmatrix}$$

$$(x-1)(-3-8) - (y-2)(-2+4) + (z-3)(7) = 0 = (x-2)(-4-10) - (y-4)(-3+5) + (z-5)(6+4)$$

$$(x-1)(-11) - (y-2)(2) + (z-3)(7) = 0 = (x-2)(-14) - (y-4)(2) + (z-5)(10)$$

$$-11x - 2y + 7z + 11 + 4 - 21 = 0 = -14x - 2y + 10z + 28 + 8 - 50$$

$$-11x - 2y + 7z - 6 = 0 = -14x - 2y + 10z - 14$$

$$11x + 2y - 7z + 6 = 0 = 7x + y - 5z + 7 \text{ is required equation.}$$

Q#4: Find the shortest distance between the lines

$$\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1} \quad \text{and} \quad \frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1}$$

Find equations of the straight line perpendicular to both the given straight lines and also its points of intersection with the given straight lines.

Solution: given lines are

$$\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1} \quad \text{--- (1)}$$

$$\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1} \quad \text{--- (2)}$$

A point on line (1) is A(3,5,7)

A point on line (2) is B(-1,-1,-1)

$$\overrightarrow{AB} = (-1-3)\hat{i} + (-1-5)\hat{j} + (-1-7)\hat{k} \Rightarrow \overrightarrow{AB} = -4\hat{i} - 6\hat{j} - 8\hat{k}$$

Here direction ratios of line (1) are 1, -2, 1 & direction ratios of line (2) are 7, -6, 1

Let $\vec{u}$ be a vector perpendicular to both lines then

$$\vec{u} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 1 \\ 7 & -6 & 1 \end{vmatrix}$$

Expanding from R_1

$$\vec{u} = (-2+6)\hat{i} - (1-7)\hat{j} + (-6+14)\hat{k}$$

$$\Rightarrow \vec{u} = 4\hat{i} + 6\hat{j} + 8\hat{k}$$

Let d be the required distance between lines then

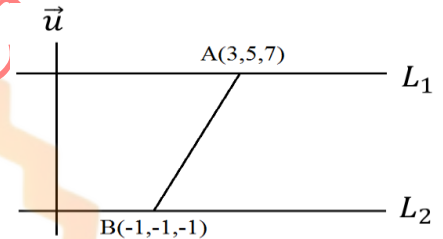
$$d = \frac{|\overrightarrow{AB} \cdot \vec{u}|}{|\vec{u}|} \Rightarrow d = \frac{(-4\hat{i} - 6\hat{j} - 8\hat{k}) \cdot (4\hat{i} + 6\hat{j} + 8\hat{k})}{\sqrt{(4)^2 + (6)^2 + (8)^2}} \Rightarrow d = \frac{-16 - 36 - 64}{\sqrt{16 + 36 + 64}} \Rightarrow d = \frac{-116}{\sqrt{116}}$$

$$d = -\sqrt{116} \Rightarrow d = -\sqrt{4 \times 29} \Rightarrow d = -2\sqrt{29} \Rightarrow d = 2\sqrt{29} \quad (\text{In magnitude})$$

Now given lines are

$$\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1} = t \quad \text{--- (1)}$$

$$\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1} = s \quad \text{--- (2)}$$



Parametric equations of given lines are

$$\left. \begin{array}{l} x = 3 + t \\ y = 5 - 2t \\ z = 7 + t \end{array} \right\} \text{---(1)} \quad \& \quad \left. \begin{array}{l} x = -1 + 7s \\ y = -1 - 6s \\ z = -1 + s \end{array} \right\} \text{---(2)}$$

Any point on line (1) is $P(3 + t, 5 - 2t, +t)$

Any point on line (2) is $Q(-1 + 7s, -1 - 6s, -1 + s)$

Direction ratios of line PQ are $3 + t + 1 - 7s, 5 - 2t + 1 + 6s, 7 + t + 1 - s$
 $= t - 7s + 4, -2t + 6s + 6, t - s + 8$

If PQ is the line of shortest then PQ is perpendicular to both lines

So

$$\begin{aligned} 1(t - 7s + 4) - 2(-2t + 6s + 6) + 1(t - s + 8) &= 0 \\ 7(t - 7s + 4) - 6(-2t + 6s + 6) + 1(t - s + 8) &= 0 \\ \begin{aligned} t - 7s + 4 + 4t - 12s - 12 + t - s + 8 &= 0 \\ 7t - 49s + 28 + 12t - 36s - 36 + t - s + 8 &= 0 \end{aligned} &\Rightarrow \begin{aligned} 6t - 20s &= 0 \\ 20t - 86s &= 0 \end{aligned} \\ \Rightarrow t = 0 \quad \& \quad s = 0 \end{aligned}$$

Hence coordinates of points P & Q are $P(3, 5, 7)$ & $Q(-1, -1, -1)$

Now equation of line of shortest distance is

$$\frac{x-3}{3+1} = \frac{y-5}{5+1} = \frac{z-7}{7+1} \Rightarrow \frac{x-3}{4} = \frac{y-5}{6} = \frac{z-7}{8} \Rightarrow \frac{x-3}{2} = \frac{y-5}{3} = \frac{z-7}{4} \text{ is required line.}$$

Q#5: Find the coordinates of the point on the join of $(-3, 7, -13)$ & $(-6, 1, -10)$ which is nearest to the intersection of the planes $2x - y - 3z + 32 = 0$ and $3x + 2y - 15z - 8 = 0$.

Solution: Equation of the line through $(-3, 7, -13)$ & $(-6, 1, -10)$ is

$$\frac{x+3}{-3+6} = \frac{y-7}{7-1} = \frac{z+13}{-13+10} \Rightarrow \frac{x+3}{3} = \frac{y-7}{6} = \frac{z+13}{-3}$$

Let $\frac{x+3}{1} = \frac{y-7}{2} = \frac{z+13}{-1} = t$

$$\left. \begin{array}{l} x = -3 + t \\ y = 7 + 2t \\ z = -13 - t \end{array} \right\} \text{---(1)}$$

Any point on line (1) is $P(-3 + t, 7 + 2t, -13 - t)$

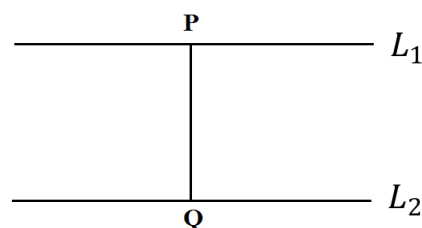
Also given equation of line is $2x - y - 3z + 32 = 0 = 3x + 2y - 15z - 8$

Let l, m, n be the direction cosines of this line

Since it lies on both planes, so by condition of perpendicularity

$$\begin{aligned} \left. \begin{array}{l} 2l - m - 3n = 0 \\ 3l + 2m - 15n = 0 \end{array} \right\} \\ \frac{l}{15+6} = \frac{-m}{-30+9} = \frac{n}{4+3} \Rightarrow \frac{l}{3} = \frac{m}{3} = \frac{n}{1} \end{aligned}$$

So direction ratios of given line are 3, 3, 1



To find a point on line , put $z = 0$ in above equations

$$\begin{cases} 2x - y + 32 = 0 \\ 3x + 2y - 8 = 0 \end{cases}$$

$$\frac{x}{8-64} = \frac{-y}{-16-96} = \frac{1}{4+3} \Rightarrow \frac{x}{-56} = \frac{-y}{-112} = \frac{1}{7} \Rightarrow x = -\frac{56}{7} = -8 \Rightarrow y = \frac{112}{7} = 16$$

So $(-8, 16, 0)$ is a point on given line

Now equation of given line through $(-8, 16, 0)$ & with direction ratios $3, 3, 1$ is

$$\frac{x+8}{3} = \frac{y-16}{3} = \frac{z}{1} = s$$

$$\begin{cases} x = -8 + 3s \\ y = 16 + 3s \\ z = s \end{cases} \quad \text{--- (2)}$$

Any point on line (2) is $Q(-8 + 3s, 16 + 3s, s)$

Now direction ratios of line PQ are $-3 + t + 8 - 3s, 7 + 2t - 16 - 3s, -13 - t - s$

$$= t - 3s + 5, 2t - 3s - 9, -t - s - 13.$$

If PQ is perpendicular to both lines (1) & (2)

Then by condition of perpendicularity

$$\begin{cases} 1(t - 3s + 5) + 2(2t - 3s - 9) - 1(-t - s - 13) = 0 \\ 3(t - 3s + 5) + 3(2t - 3s - 9) + 1(-t - s - 13) = 0 \end{cases} \Rightarrow \begin{cases} t - 3s + 5 + 4t - 6s - 18 + t + s + 13 = 0 \\ 3t - 9s + 15 + 6t - 9s - 27 - t - s - 13 = 0 \end{cases}$$

$$6t - 8s = 0 \quad \text{--- (I)}$$

$$8t - 19s - 25 = 0 \quad \text{--- (II)}$$

Multiplying (I) by 4 & (II) by 3 and subtracting (II) from (I)

$$25s = -75 \Rightarrow s = -3$$

$$\text{Put in (I)} \quad 6t - 8(-3) = 0 \Rightarrow 6t + 24 = 0 \Rightarrow t + 4 = 0 \Rightarrow t = -4$$

$$\text{Put } t = -4 \text{ in coordinates of } P(-3 - 4, 7 - 8, -13 + 4) \Rightarrow P(-7, -1, -9)$$

Q#6: Find the length and equations of the common perpendicular of the lines

$$L : 6x + 8y + 3z - 13 = 0, \quad x + 2y + z - 3 = 0$$

$$M : 3x - 9y + 5z = 0, \quad x + y - z = 0$$

Solution: given lines are

$$L : 6x + 8y + 3z - 13 = 0, \quad x + 2y + z - 3 = 0$$

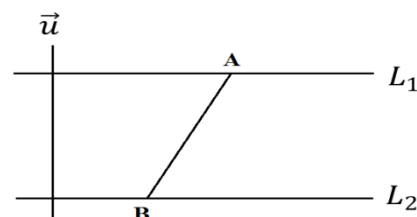
$$M : 3x - 9y + 5z = 0, \quad x + y - z = 0$$

We will write both equations in symmetric form.

Let l_1, m_1, n_1 be direction cosines of line L. Since it lies on both planes. Hence by condition of perpendicularity

$$\begin{cases} 6l_1 + 8m_1 + 3n_1 = 0 \\ l_1 + 2m_1 + n_1 = 0 \end{cases}$$

$$\frac{l_1}{8-6} = \frac{-m_1}{6-3} = \frac{n_1}{12-8} \Rightarrow \frac{l_1}{2} = \frac{m_1}{-3} = \frac{n_1}{4}$$



So direction ratios of line L are 2,-3,4

To find a point on line L, put $z = 0$

$$\begin{cases} 6x + 8y - 13 = 0 \\ x + 2y - 3 = 0 \end{cases}$$

$$\frac{x}{-24 + 26} = \frac{-y}{-18 + 13} = \frac{1}{12 - 8} \Rightarrow \frac{x}{2} = \frac{y}{5} = \frac{1}{4} \Rightarrow x = \frac{1}{2}, \quad y = \frac{5}{4}$$

So a point on line L is $\left(\frac{1}{2}, \frac{5}{4}, 0\right)$

Now equation of line L through $\left(\frac{1}{2}, \frac{5}{4}, 0\right)$ & having direction ratios 2,-3,4 is

$$\frac{x - 1/2}{2} = \frac{y - 5/4}{-3} = \frac{z}{4} \quad \text{--- (1)}$$

Next suppose l_2, m_2, n_2 be the direction cosines of line M. Since it lies on both planes, so by condition of perpendicularity

$$\begin{cases} 3l_2 - 9m_2 + 5n_2 = 0 \\ l_2 + m_2 - n_2 = 0 \end{cases}$$

$$\frac{l_2}{9 - 5} = \frac{-m_2}{-3 - 5} = \frac{n_2}{3 + 9} \Rightarrow \frac{l_2}{4} = \frac{m_2}{8} = \frac{n_2}{12} \Rightarrow \frac{l_2}{1} = \frac{m_2}{2} = \frac{n_2}{3}$$

So direction ratios of line M are 1,2,3

To find a point on line M, put $z = 0$

$$\begin{cases} 3x - 9y = 0 \\ x + y = 0 \end{cases} \Rightarrow \begin{cases} x - 3y = 0 \quad \text{--- (I)} \\ -x + y = 0 \quad \text{--- (II)} \end{cases}$$

Subtracting we have

$$-4y = 0 \Rightarrow y = 0$$

Put in (I)

$$x - 0 = 0 \Rightarrow x = 0$$

Now a point on line M is (0,0,0)

Hence equation of line M through (0,0,0) having direction ratios 1,2,3

$$\frac{x}{1} = \frac{y}{2} = \frac{z}{3} \quad \text{--- (2)}$$

Now we want to find shortest distance between lines (1) & (2)

A point on line (1) is $A\left(\frac{1}{2}, \frac{5}{4}, 0\right)$

A point on line (2) is $B(0,0,0)$

$$\overrightarrow{AB} = -\frac{1}{2}\hat{i} - \frac{5}{4}\hat{j} + 0\hat{k}$$

Let $\vec{u}$ be a vector perpendicular to both lines then
$$\vec{u} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -3 & 4 \\ 1 & 2 & 3 \end{vmatrix}$$

Expanding from R_1

$$\vec{u} = (-9 - 8)\hat{i} - (6 - 4)\hat{j} + (4 + 3)\hat{k} \Rightarrow \vec{u} = -17\hat{i} - 2\hat{j} + 7\hat{k}$$

Suppose d be the required shortest distance between lines then

$$d = \frac{\overrightarrow{AB} \cdot \vec{u}}{|\vec{u}|} = \frac{\left(-\frac{1}{2}\hat{i} - \frac{5}{4}\hat{j} + 0\hat{k}\right) \cdot (-17\hat{i} - 2\hat{j} + 7\hat{k})}{\sqrt{(-17)^2 + (-2)^2 + (7)^2}} \Rightarrow d = \frac{\frac{17}{2} + \frac{5}{2} + 0}{\sqrt{289 + 4 + 49}} \Rightarrow d = \frac{22/2}{\sqrt{342}} \Rightarrow d = \frac{11}{\sqrt{342}}m$$

Now equation of common perpendicular is

$$\begin{vmatrix} x - \frac{1}{2} & y - \frac{5}{4} & z \\ 2 & -3 & 4 \\ -17 & -2 & 7 \end{vmatrix} = 0 = \begin{vmatrix} x & y & z \\ 1 & 2 & 3 \\ -17 & -2 & 7 \end{vmatrix}$$

$$\left(x - \frac{1}{2}\right)(-21 + 8) - \left(y - \frac{5}{4}\right)(14 + 68) + z(-4 - 51) = 0 = x(14 + 6) - y(7 + 51) + z(-2 + 34)$$

$$\left(x - \frac{1}{2}\right)(-13) - \left(y - \frac{5}{4}\right)(82) + z(-55) = 0 = 20x - 58y + 32z$$

$$-13x - 82y - 55z + \frac{13}{2} + \frac{205}{2} = 0 = 20x - 58y + 32z$$

$$-13x - 82y - 55z + \frac{218}{2} = 0 = 10x - 29y + 16z$$

$$-13x - 82y - 55z + 109 = 0 = 10x - 29y + 16z$$

$$13x + 82y + 55z - 109 = 0 = 10x - 29y + 16z \text{ is required line}$$

Q#7: Show that the shortest distance between ant two opposite edges of the tetrahedron formed by the planes $y + z = 0$, $z + x = 0$, $x + y = 0$, $x + y + z = a$ is $\frac{2a}{\sqrt{6}}$ and that the three straight lines of the shortest distances intersect at the point $(-a, -a, -a)$.

Solution: Suppose the planes $y + z = 0$, $z + x = 0$, $x + y = 0$ & $x + y + z = a$ be represented by ABC, ACD, ABD & BCD respectively.

The equation of line AC is

$$\begin{cases} y + z = 0 \\ z + x = 0 \end{cases} \text{ or } y = -z \text{ \& } x = -z$$

$$\text{or } \frac{x}{1} = \frac{y}{1} = \frac{z}{-1} \text{ --- (1) is symmetric form of AC.}$$

Now the equation of opposite edges BD is

$$\begin{cases} x + y = 0 \\ x + y + z = a \end{cases}$$

Let l, m, n be the direction cosines of this line.

Since it lies on both planes. So by condition of perpendicularity

$$\begin{cases} l + m + 0n = 0 \\ l + m + n = 0 \end{cases}$$

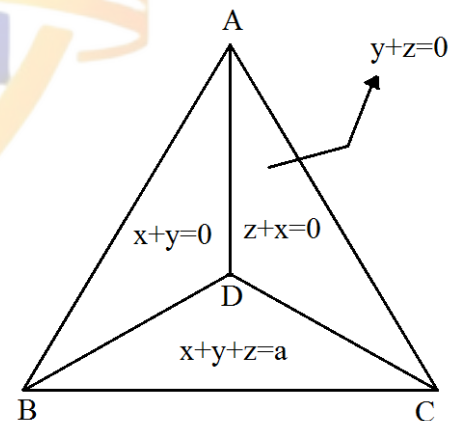
$$\frac{l}{1-0} = \frac{-m}{1-0} = \frac{n}{1-1} \Rightarrow \frac{l}{1} = \frac{m}{-1} = \frac{n}{0}$$

So direction ratios of line is $1, -1, 0$

To find a point on this line, put $x = 0$ in above equations $\begin{cases} 0 + y = 0 \\ 0 + y + z = a \end{cases} \text{ or } y = 0 \text{ \& } y + z = a \Rightarrow z = a$

So a point on this line BD is $(0, 0, a)$, Hence equation of this line is

$$\frac{x}{1} = \frac{y}{-1} = \frac{z-a}{0} \text{ --- (2)}$$



Now we will find shortest distance between line (1) & (2)

A point on line (1) is $A(0,0,0)$

A point on line (2) is $B(0,0,a)$

Now $\vec{AB} = 0\hat{i} + 0\hat{j} + a\hat{k}$

Let $\vec{u}$ be a vector perpendicular to both lines then

$$\vec{u} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 1 & -1 & 0 \end{vmatrix}$$

Expanding from R_1 $\vec{u} = (0-1)\hat{i} - (0+1)\hat{j} + (-1-1)\hat{k} \Rightarrow \vec{u} = -\hat{i} - \hat{j} - 2\hat{k}$

Let d be the shortest distance between lines then

$$d = \frac{|\vec{AB} \cdot \vec{u}|}{|\vec{u}|} = \frac{(0\hat{i} + 0\hat{j} + a\hat{k}) \cdot (-\hat{i} - \hat{j} - 2\hat{k})}{\sqrt{1+1+4}} \Rightarrow d = \frac{0+0-2a}{\sqrt{6}} \Rightarrow d = \frac{2a}{\sqrt{6}} \text{ is required distance.}$$

Similarly we can show that the shortest distance between opposite edges AB,CD & BC,AD is also $\frac{2a}{\sqrt{6}}$

Now equation of line of shortest distance between opposite edges AC & BD is

$$\begin{vmatrix} x & y & z \\ 1 & 1 & -1 \\ -1 & -1 & -2 \end{vmatrix} = 0 = \begin{vmatrix} x & y & z-a \\ 0 & -1 & 0 \\ -1 & -1 & -2 \end{vmatrix}$$

$$x(-2-1) - y(-2-1) + z(-1+1) = 0 = x(2-0) - y(-2+0) + (z-a)(-1-1)$$

$$x(-3) - y(-3) + z(0) = 0 = x(2) - y(-2) + (z-a)(-2)$$

$$-3x + 3y + 0z = 0 = 2x + 2y - 2z + 2a$$

$$x - y = 0 = x + y - z + a$$

We see that the point $(-a, -a, -a)$ satisfies this equation, So this point lies on the line of shortest distance between AC & BD. Similarly $(-a, -a, -a)$ also lies on the other two lines of shortest distance.

Hence it lies on the intersection of all three lines of shortest distance.

Q#8: Find the shortest distance between the straight line joining the points $A(3, 2, -4)$ & $B(1, 6, -6)$ and the straight line joining the points $C(-1, 1, -2)$ & $D(-3, 1, -6)$. Also find equation of the line of shortest distance and coordinates of the feet of the common perpendicular.

Solution: Equation of line passing through $A(3,2,-4)$ & $B(1,6,-6)$ is

$$\frac{x-3}{1-3} = \frac{y-2}{6-2} = \frac{z+4}{-6+4} \Rightarrow \frac{x-3}{-2} = \frac{y-2}{4} = \frac{z+4}{-2} \Rightarrow \frac{x-3}{1} = \frac{y-2}{-2} = \frac{z+4}{1} \quad \text{--- (1)}$$

& equation of line through $C(-1,1,-2)$ & $D(-3,1,-6)$ is

$$\frac{x+1}{-3+1} = \frac{y-1}{1-1} = \frac{z+2}{-6+2} \Rightarrow \frac{x+1}{-2} = \frac{y-1}{0} = \frac{z+2}{-4} \Rightarrow \frac{x+1}{1} = \frac{y-1}{0} = \frac{z+2}{2} \quad \text{--- (2)}$$

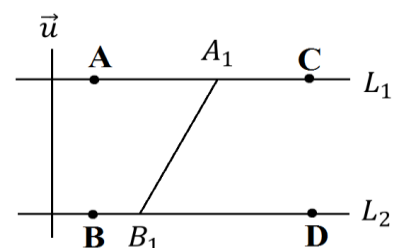
A point on line (1) is $A_1(3,2,-4)$

A point on line (2) is $B_1(-1,1,-2)$

$$\vec{A_1B_1} = (-1-3)\hat{i} + (1-2)\hat{j} + (-2+4)\hat{k} \Rightarrow \vec{A_1B_1} = -4\hat{i} - \hat{j} + 2\hat{k}$$

Let $\vec{u}$ be a vector perpendicular to both lines (1) & (2) then

$$\vec{u} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 1 \\ 1 & 0 & 2 \end{vmatrix}$$



Expanding by R_1

$$\vec{u} = (-4 - 0)\hat{i} - (2 - 1)\hat{j} + (0 + 2)\hat{k} \Rightarrow \vec{u} = -4\hat{i} - \hat{j} + 2\hat{k}$$

Let d be the required shortest distance between lines (1) & (2) then

$$d = \frac{|\vec{A_1B_1} \cdot \vec{u}|}{|\vec{u}|} = \frac{(-4\hat{i} - \hat{j} + 2\hat{k}) \cdot (-4\hat{i} - \hat{j} + 2\hat{k})}{\sqrt{16+1+4}} \Rightarrow d = \frac{16+1+4}{\sqrt{21}} \Rightarrow d = \frac{21}{\sqrt{21}} \Rightarrow d = \sqrt{21}$$

As lines are

$$\frac{x-3}{1} = \frac{y-2}{-2} = \frac{z+4}{1} = t \quad \text{--- (1)}$$

$$\& \quad \frac{x+1}{1} = \frac{y-1}{0} = \frac{z+2}{2} = s \quad \text{--- (2)}$$

Any point on line (1) is $P(3+t, 2-2t, -4+t)$

Any point on line (2) is $Q(-1+s, 1, -2+2s)$

Direction ratios of PQ are $3+t+1-s, 2-2t-1, -4+t+2-2s$

Direction ratios of line PQ are $t-s+4, -2t+1, t-2s-2$

Suppose PQ is line of shortest distance then PQ is perpendicular to both lines (1) & (2)

So by condition of perpendicularity

$$\begin{aligned} 1(t-s+4) - 2(-2t+1) + 1(t-2s-2) &= 0 \\ 1(t-s+4) + 0(-2t+1) + 2(t-2s-2) &= 0 \\ \left. \begin{aligned} t-s+4+4t-2+t-2s-2 &= 0 \\ t-s+4+2t-4s-4 &= 0 \end{aligned} \right\} &\Rightarrow \begin{aligned} 6t-5s &= 0 \\ 3t-5s &= 0 \end{aligned} \Rightarrow t=0 \& \ s=0 \end{aligned}$$

So coordinates of feet of perpendicular P & Q are $P(3, 2, -4)$ & $Q(-1, 1, -2)$.

Now equation of common perpendicular PQ is

$$\frac{x-3}{-1-3} = \frac{y-2}{1-2} = \frac{z+4}{-2+4} \Rightarrow \frac{x-3}{-4} = \frac{y-2}{-1} = \frac{z+4}{2} \quad \text{or} \quad \frac{x-3}{4} = \frac{y-2}{1} = \frac{z+4}{-2}$$

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Muhammad Umar Asghar sb (M.Sc Mathematics)

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